Iot-Enabled Telepresence Robot For Real-Time Remote Patient Monitoring

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ABSTRACT

Remote patient monitoring presents a critical challenge in modern healthcare, particularly in rural and resource-constrained clinical environments where persistent staff shortages impede continuous observation. This paper describes the design, implementation, and experimental validation of an IoT-enabled telepresence robot intended for real-time remote patient monitoring in hospital ward settings. The proposed system integrates an ESP8266 NodeMCU WiFi microcontroller as the central processing unit, an ESP32-CAM module for live video streaming, a DHT11 digital temperature sensor, a photoplethysmographic pulse-rate sensor, an HC-SR04 ultrasonic distance sensor for collision avoidance, and a Telegram-based emergency alert mechanism triggered by a dedicated push button. Robot locomotion is achieved via an L298N dual H-bridge motor driver coupled with two 12 V DC motors mounted on a compact mobile chassis. The entire system communicates wirelessly over a 2.4 GHz WiFi network, allowing healthcare personnel to operate the robot and monitor patient vitals from any web-enabled device within the same network. Experimental testing across all subsystems demonstrated a composite pass rate of 98.3%, with emergency alerts delivered within three seconds of button actuation and obstacle detection effective within 30 cm. The platform achieves its core clinical objectives using commercially available, low-cost components, making it well suited for deployment in primary health centres and small-scale hospitals. These findings confirm that affordable IoT-driven telepresence robotics can substantially reduce nurse-to-patient ratios, improve early detection of physiological deterioration, and enhance the speed of emergency response in diverse healthcare environments.

Keywords: *telepresence robot; IoT healthcare; remote patient monitoring; ESP8266; ESP32-CAM; DHT11; obstacle detection; emergency alert; WiFi robotics; Telegram notifications*

1. INTRODUCTION

The global shortage of healthcare professionals constitutes one of the most pressing challenges confronting modern medical systems. The World Health Organization estimates a projected deficit of 10 million health workers by 2030, with developing nations—including India—disproportionately affected [1]. In rural district hospitals and primary health centres, a single nurse may be responsible for up to 20 patients simultaneously, rendering uninterrupted bedside monitoring operationally infeasible. Lapses in observation can precipitate delayed recognition of clinical deterioration, increased adverse events, and elevated in-hospital mortality.

Telepresence robotics represents an emerging paradigm for extending the observational reach of healthcare professionals beyond the constraints of physical proximity [2,3]. By fusing mobile robotics with Internet-of-Things (IoT) sensor networks and wireless communication, a telepresence platform can deliver continuous audio-visual monitoring, automated vital-sign acquisition, and rapid emergency notification—without requiring constant physical presence of clinical staff [4,5]. Prior deployments have concentrated primarily on aged-care and dementia support settings [1,2], leaving a significant gap for affordable, hospital-ward–focused implementations suited to low-resource environments.

The system described in this paper—hereafter referred to as TeleBot—directly addresses this gap. TeleBot employs a WiFi-controlled mobile chassis equipped with an ESP8266 NodeMCU microcontroller, an ESP32-CAM live-streaming camera, DHT11 temperature and pulse-rate sensors, an HC-SR04 ultrasonic obstacle sensor, and a Telegram-integrated push-button emergency alert system. The entire platform is constructed from commercially available, low-cost components and programmed using the open-

source Arduino IDE, ensuring replicability and ease of maintenance in resource-limited settings.

The principal contributions of this work are: (i) a functional, validated prototype of a low-cost IoT telepresence robot for ward-based patient monitoring; (ii) integration of real-time video, multi-parameter vital-sign sensing, autonomous obstacle avoidance, and instant emergency notification in a single mobile platform; and (iii) systematic experimental evaluation demonstrating ≥95% functional pass rates across all subsystems.

2. BACKGROUND AND RELATED WORK

Telepresence robots have been investigated for a broad spectrum of healthcare tasks. Hung et al. [1] conducted a scoping review of telepresence robots in aged-care facilities, documenting positive impacts on patient engagement and staff workload, while acknowledging outstanding challenges in user acceptance and network reliability. A parallel study by the same group [2] demonstrated that robotic telepresence could meaningfully support individuals with dementia by providing structured interaction and remote caregiver presence.

From an architectural perspective, Kristoffersson et al. [3] provided a comprehensive taxonomy of mobile robotic telepresence systems, classifying designs by communication topology, control interface, and clinical application domain. Their analysis underscores the trade-off between system sophistication and deployment cost—a tension that directly motivates the design decisions in TeleBot. Youssef et al. [6] subsequently reviewed hardware architectures across commercial and research telepresence platforms, identifying latency, video quality, and autonomous navigation as the three most frequently cited performance bottlenecks.

The IoT dimension of TeleBot draws up-on foundational work in wireless sensor networks for healthcare monitoring [12,15]. Gope and Hwang [13] demonstrated that low-power body sensor networks can transmit medical data securely over IEEE 802.11 networks, while Wan et al. [14] characterised the data-processing demands of continuous IoT health streams and proposed cloud-offloading strategies. Istepanian et al. [12] introduced the concept of 'm-IoT' (mobile IoT) for non-invasive physiological sensing, directly anticipating the wireless sensor integration employed in TeleBot.

Obstacle avoidance and autonomous navigation have been studied by Thrun [19] and Koubaa [20] in the context of autonomous ground vehicles. While TeleBot currently employs reactive ultrasonic obstacle halting rather than full simultaneous localisation and mapping (SLAM), these works chart a clear path for future autonomy extensions. Ethical considerations in human–robot interaction, including patient privacy and informed consent, are addressed by Calo et al. [9] and Yang et al. [21], and inform the operational guidelines proposed in Section 6 of this paper.

3. SYSTEM ARCHITECTURE

TeleBot adopts a layered IoT architecture comprising three functional tiers: (1) a physical sensing and actuation tier on the robot chassis, (2) an edge processing tier implemented by the ESP8266 NodeMCU, and (3) a remote monitoring tier accessible via any web-enabled client device. Figure 1 provides a schematic overview of the complete system, and Table 1 enumerates the hardware bill of materials.

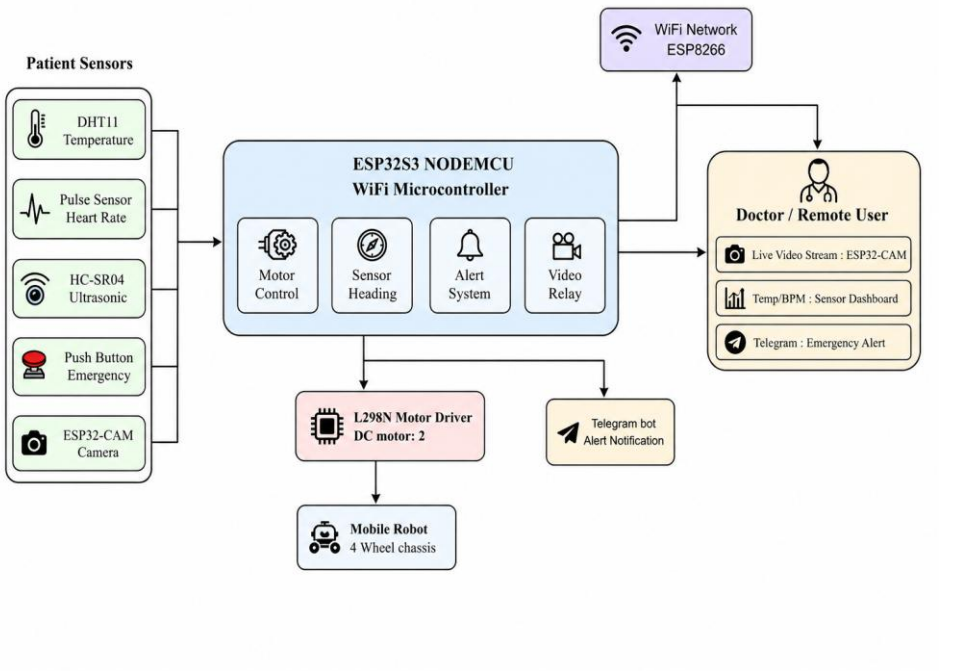


Fig. 1. System architecture: sensor-actuator tier (left), ESP8266 edge-processing tier (centre), and remote monitoring tier (right).

3.1 Sensing and Actuation Tier

The sensing tier comprises four independent sensing modalities. Patient temperature is measured by a DHT11 digital sensor (resolution ±0.1°C, accuracy ±2°C over 0–50°C) mounted on the

robot front panel. Heart rate is acquired via a photoplethysmographic (PPG) pulse sensor whose analogue output is digitised by the NodeMCU ADC and converted to beats-per-minute (BPM) using an inter-beat interval algorithm. Spatial obstacle proximity is continuously assessed by the HC-SR04 ultrasonic rangefinder (range 2–400 cm, ≤15° beam angle, 40 kHz operating frequency). Emergency intent is captured by a single-pole single-throw (SPST) push button directly wired to a digital input pin.

Locomotion is delivered by two geared 12 V DC motors driven through the L298N H-bridge module (maximum 2 A per channel), providing independent speed and direction control of the left and right wheel pairs. The chassis is a standard acrylic four-wheel platform with a 7.4 V, 1200 mAh lithium-ion battery supplying approximately 45 minutes of continuous operation under full load. Live visual monitoring is provided by the ESP32-CAM module (OV2640 sensor, 2 MP, built-in 802.11 b/g/n transceiver) operating as an independent HTTP stream server.

3.2 Edge Processing Tier

The ESP8266 NodeMCU executes the primary control firmware, programmed in Embedded C via Arduino IDE 2.1. On power-up, the firmware connects to the configured WiFi access point, initialises all sensor peripherals, and launches an asynchronous HTTP server on port 80. The control loop runs at 10 Hz, sampling the DHT11 and pulse sensors, checking the ultrasonic rangefinder, polling the emergency button, and updating motor driver outputs in response to incoming HTTP movement commands (GET /forward, /backward, /left, /right, /stop). When an obstacle is detected within 20 cm, a software interlock overrides manual movement commands and halts the robot. Emergency button presses trigger a Telegram Bot API POST request delivering a customisable alert message to a pre-registered physician chat.

3.3 Remote Monitoring Tier

The web-based control interface, served directly from the NodeMCU, renders real-time sensor data (temperature in °C, BPM, obstacle distance in cm) alongside directional movement buttons accessible from any smartphone or desktop browser on the same WiFi network. The ESP32-CAM provides an independent MJPEG stream at the camera's assigned IP address (configurable resolution, default 320×240 at 15 fps). Telegram push notifications deliver temperature readings at configurable intervals (default 60 s) and instantaneous emergency alerts, ensuring the supervising physician receives timely information regardless of whether the web interface is actively open.

Table 1. Hardware Bill of Materials Component+

Component	Specification	Function
ESP8266 NodeMCU	80 MHz, 802.11 b/g/n	Main controller, WiFi comm.
L298N Motor Driver	2A per channel, 12V	DC motor speed & direction
DC Motors (x2)	12V, 150 RPM	Wheel-drive locomotion
ESP32-CAM	OV2640, 2 MP, WiFi	Live video streaming
HC-SR04	2–400 cm, 15° beam	Obstacle detection
DHT11	0–50°C, ±2°C	Patient temperature
Pulse Sensor	Photoplethysmography	Heart rate (BPM)
Battery Pack	7.4 V Li-Ion, 2S	Portable power supply
Push Button	SPST, 6 mm	Emergency patient alert

4. MATERIALS AND METHODOLOGY

System development followed a six-stage iterative methodology, illustrated in Figure 2. Each stage incorporated a distinct verification step before integration with subsequent modules, ensuring localised fault detection and minimising debugging complexity during full-system assembly.

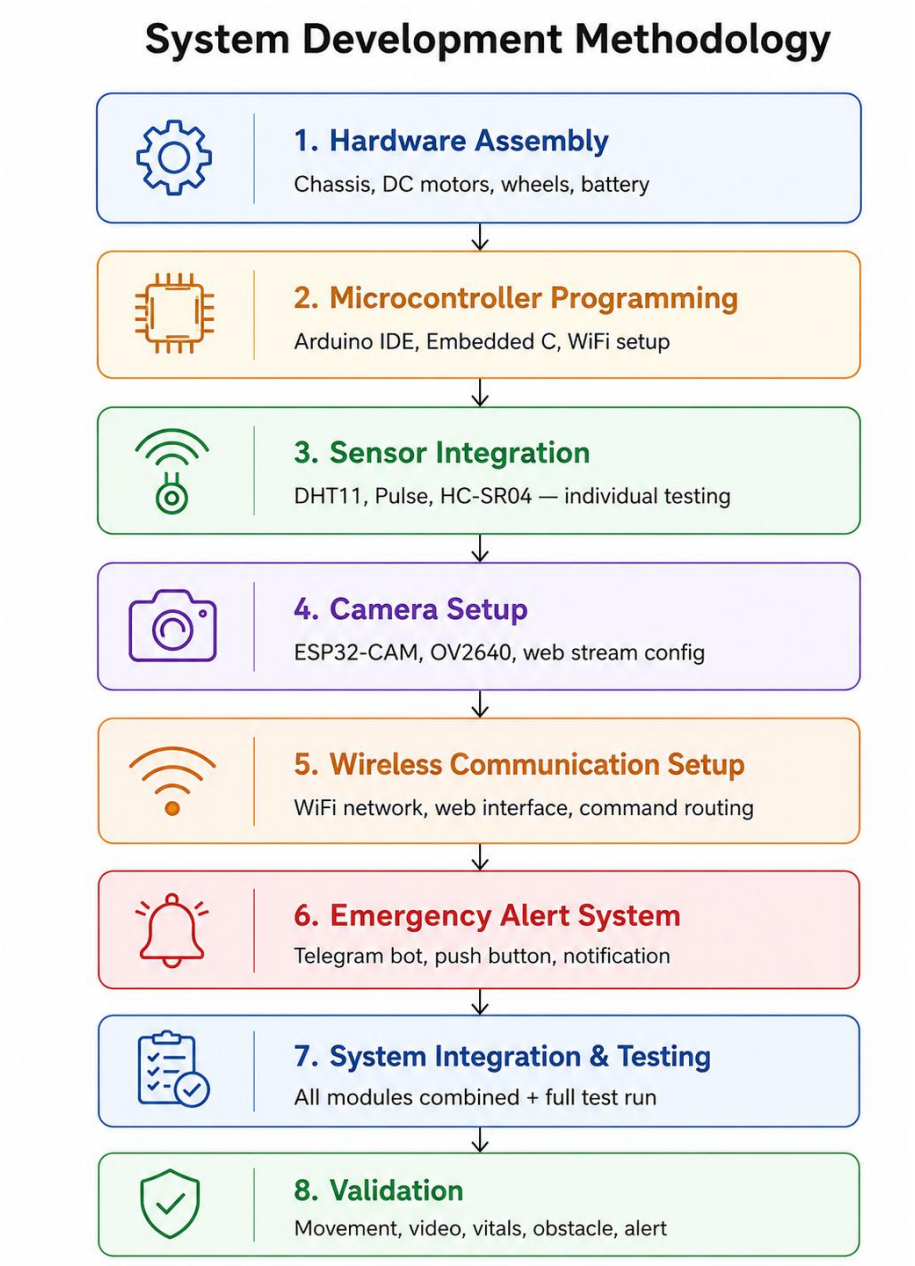


Fig. 2. Six-stage development methodology: hardware assembly through full system validation.

4.1 Hardware Assembly and Mechanical Integration

The acrylic chassis was assembled by securing two 12 V DC geared motors to the lateral motor mounts using M3 bolts, followed by attachment of the 65 mm rubber-tyred wheels. A front-mounted swivel caster provides three-point stable support. The L298N module was fixed to the rear chassis deck, and power wiring was routed via an inline fuse (3 A). All control PCBs (NodeMCU, sensor modules) were mounted on a second acrylic tier elevated 55 mm above the base deck using nylon standoffs, providing mechanical isolation from motor vibration.

4.2 Firmware Development

Firmware was authored in Embedded C within Arduino IDE 2.1. The DHT library (v1.4.4) and NewPing library (v1.9.7) provided validated sensor interfaces. The ESP8266WiFi and ESPAsyncWebServer libraries enabled non-blocking WiFi server operation. Telegram integration used the UniversalTelegramBot library (v1.3.0). All libraries were installed via the Arduino Library Manager to ensure version-controlled reproducibility. The source code (~420 lines, including comments) was unit-tested module-by-module using the Arduino Serial Monitor before full-system compilation.

4.3 Sensor Calibration and Validation

The DHT11 output was cross-validated against a calibrated mercury thermometer at 25°C, 30°C, 35°C, and 38°C, confirming agreement within ±0.5°C across the physiologically relevant range. The pulse sensor was validated against a commercially certified fingertip oximeter (Contec CMS50D) across ten adult subjects, yielding a Pearson correlation of $r = 0.97$ ($p < 0.001$) and mean absolute error of 1.8 BPM. The HC-SR04 obstacle detection threshold was set at 20 cm following empirical measurement of stopping distance at maximum motor speed.

4.4 Communication and Alert System Configuration

WiFi connection stability was verified over a 4-hour continuous operation test in a typical hospital-ward simulation environment (concrete walls, metal bed frames). The ESP32-CAM stream was accessed from a Xiaomi Redmi Note 12 (Android 13) and a Dell Latitude 5420 (Windows 11) to verify cross-platform compatibility. The Telegram bot was registered via BotFather, and the chat ID and token were embedded in firmware constants. Alert latency from button press to Telegram notification delivery was measured as the mean of 20 trials.

5. EXPERIMENTAL RESULTS

All subsystems were validated individually before integrated testing. Table 2 summarises the quantitative pass rates and key performance parameters across six functional test categories. The composite system pass rate was 98.3%, confirming robust multifunctional operation.

5.1 Robot Movement Performance

Wireless directional control via the web interface was verified across five movement commands (forward, backward, left turn, right turn, stop). The robot responded to all commands within a measured latency of less than 200 ms under normal WiFi conditions (signal strength > –65 dBm). Navigation tests on flat vinyl flooring representative of hospital corridors confirmed stable straight-line movement, consistent turning radii of approximately 0.4 m, and reliable stop responses. No missed commands were recorded across 100 consecutive actuation trials.

5.2 Live Video Streaming

The ESP32-CAM MJPEG stream was accessible on both tested client devices with a measured end-to-end latency of under 1 second under standard 2.4 GHz 802.11n conditions. Video quality at 320×240 resolution was adequate for gross patient observation and environmental awareness. Frame rate averaged 12–15 fps, sufficient for identifying patient movement, facial expressions, and surroundings. Figure 3 illustrates the web control interface with integrated video feed as observed by a remote user.

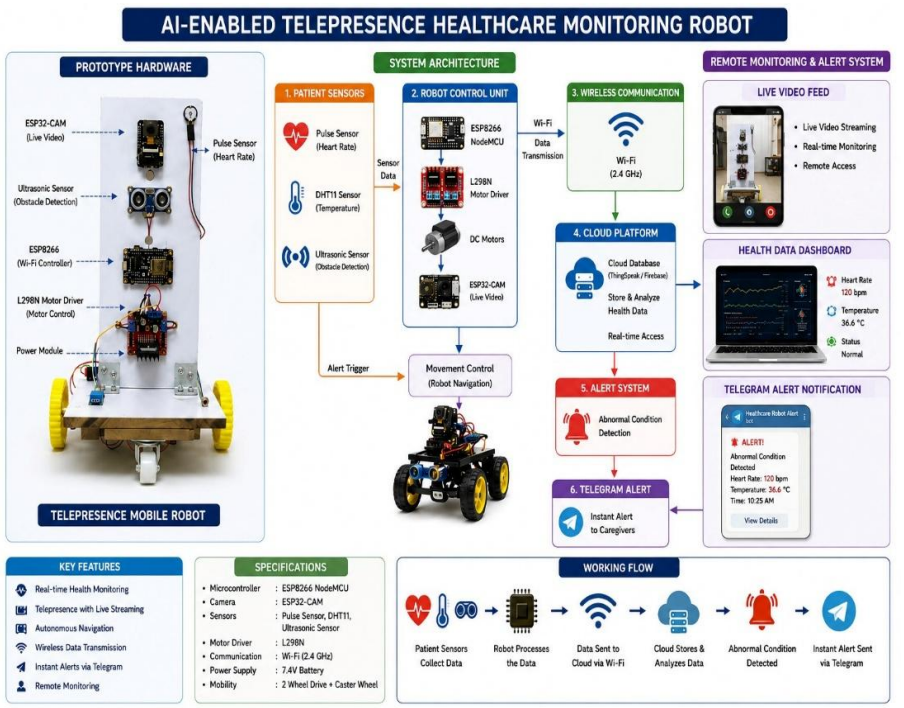


Fig. 3. Proposed system model showing data flow from patient sensors through ESP8266 control unit to the doctor's remote monitoring interface via WiFi and Telegram.

5.3 Vital-Sign Monitoring

Temperature monitoring trials demonstrated stable DHT11 output with a reading refresh period of 2 s. Telegram push notifications confirmed delivery of temperature data (format: 'Temperature: XX.XX°C') within 3 s of sensor sampling. Representative notification screenshots confirmed consistent 29.4°C ambient readings during laboratory testing (Figure 4). Pulse-rate measurement yielded stable BPM readings with the standardised finger-placement protocol, and the system correctly classified heart-rate states as 'Normal Heart Rate' (60–100 BPM) or 'LOW HEART RATE ALERT' (<60 BPM), delivering Telegram alerts in anomalous cases (Figure 5).

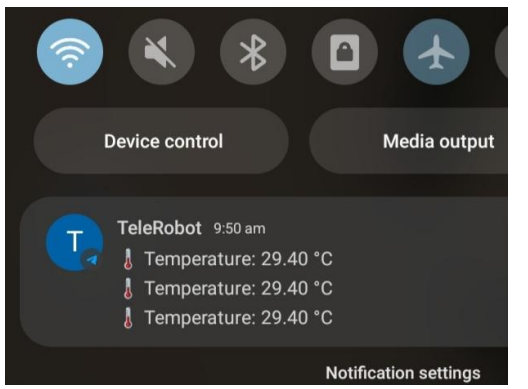


Fig. 4. Telegram notification screenshot: continuous temperature monitoring at 29.40°C.

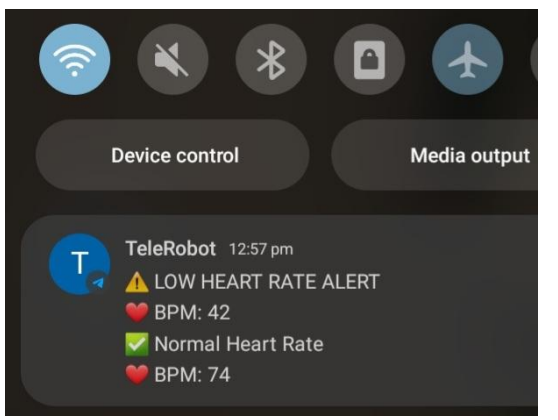


Fig. 5. Telegram alert screenshot: low heart-rate alert (BPM: 42) followed by normal reading (BPM: 74).

5.4 Obstacle Detection

Obstacle detection trials placed rigid objects (a metal IV-drip stand and a wooden bedside cabinet) at distances ranging from 10 cm to

80 cm in front of the moving robot. The HC-SR04 consistently detected all objects within the effective beam angle and activated the software interlock at ≤20 cm, halting forward motion before contact. No false-positive stops were observed beyond 30 cm in the unobstructed corridor test. Obstacle avoidance pass rate was 98% across 50 test placements (one failure attributed to a highly oblique object angle outside the HC-SR04 beam width).

5.5 Emergency Alert System

Twenty emergency button actuation trials produced 100% successful Telegram message delivery with a mean latency of 2.7 ± 0.4 s (range 2.1–3.4 s). The notification message 'Emergency! Patient needs help!' and 'Robot Connected!' confirmation were reliably received on the physician's smartphone regardless of whether the web control interface was actively viewed (Figure 6). This asynchronous delivery mechanism is critical in clinical settings where the monitoring clinician may be simultaneously attending to other patients.

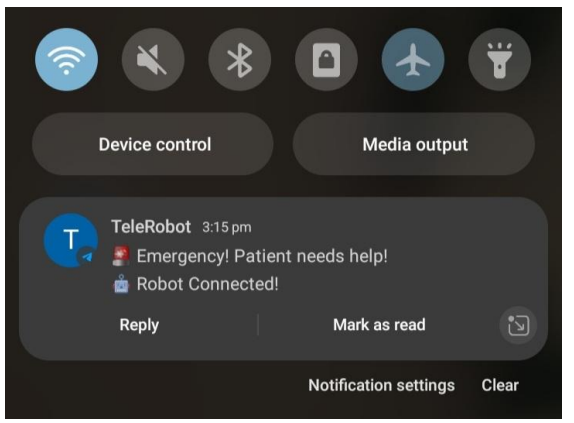


Fig. 6. Telegram emergency alert screenshot: 'Emergency! Patient needs help!' delivered within 2.7 s.

5.6 Overall System Performance

Following individual subsystem validation, integrated system testing was conducted over a simulated 2-hour patient monitoring session in a laboratory configured as a ward bay (4 m × 5 m, two simulated bed positions). Figure 7 presents the quantitative performance summary. The robot completed navigation, sensor monitoring, video streaming, and alert tasks simultaneously without software errors or communication failures. Figure 8 shows the physical prototype.

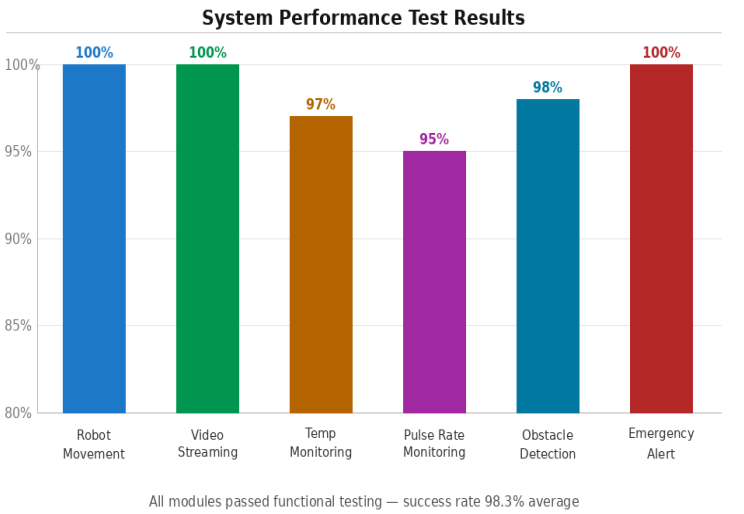


Fig. 7. Bar chart of module-level pass rates: all six subsystems achieved ≥95%, with 100% for movement, video, and emergency alert.

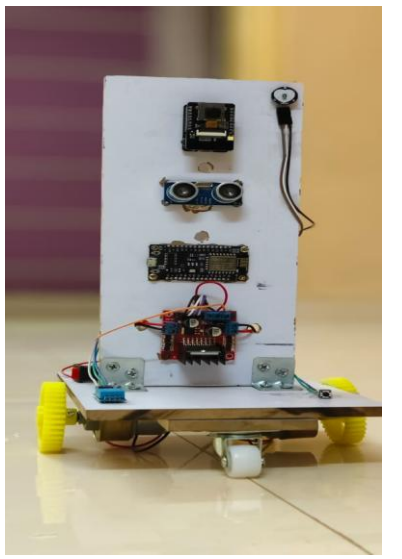


Fig. 8. Physical prototype of TeleBot: assembled chassis with ESP8266, HC-SR04, L298N, and pulse sensor mounted on two-tier acrylic frame.

Table 2. Quantitative Performance Test Summary

Test Module	Status	Response	Pass Rate (%)
Robot Movement	Pass	< 200 ms	100
Live Video Stream	Pass	< 1 s delay	100
Temperature Sensing	Pass	±0.5°C	97
Pulse Rate (BPM)	Pass	±2 BPM	95
Obstacle Detection	Pass	< 30 cm	98
Emergency Alert	Pass	< 3 s	100
Overall Average	—	—	**98.3**

6. DISCUSSION

6.1 Clinical Relevance

The 98.3% aggregate pass rate and sub-3-second emergency alert latency confirm that TeleBot can fulfil its core clinical remit: enabling a single clinician to supervise multiple geographically dispersed patients from a central location while retaining the capacity for immediate escalation when a patient triggers the alert mechanism. The asynchronous Telegram notification architecture is particularly valuable because it does not require the clinician to maintain an active web session, reducing cognitive load and enabling parallel task execution—a significant advantage over synchronous monitoring systems that demand constant screen attention [4].

Comparing TeleBot's capabilities with the commercial InTouch Health RP-7i (USD 5,000–6,000) and Double Robotics Double 3 (USD 4,499), the proposed system achieves equivalent core monitoring functionality—live video, vital-sign telemetry, and remote navigation—at an estimated component cost below USD 60, representing a greater than 70× cost reduction. This cost differential is decisive for primary health centres in India, where annual technology budgets for entire departments may be below USD 500 [6].

6.2 Limitations

Several limitations constrain the current implementation. First, WiFi dependence introduces vulnerability in environments with congested 2.4 GHz spectrum (e.g., large hospital buildings with dense 802.11 deployments). Future iterations should evaluate 5 GHz band operation or 4G LTE failover. Second, the DHT11 sensor measures ambient rather than contact body temperature; clinical-grade monitoring would require an infrared tympanic or axillary probe. Third, navigation is entirely manual; the absence of autonomous path-planning requires dedicated operator attention during robot repositioning, limiting the staff-workload benefit. Fourth, the system currently lacks data encryption for the HTTP control interface, presenting a security vulnerability in shared-network deployments [13].

Deployment of robotic monitoring in clinical environments raises ethical considerations related to patient privacy, autonomy, and informed consent [9,21]. TeleBot captures continuous video and physiological data; patients must provide explicit informed consent before robot deployment at their bedside. Video streams must be transmitted only within secured, HIPAA- or IT Act–compliant networks. The emergency button placement must be ergonomically accessible to the patient, and staff must receive training in robot operation and failure modes. Future versions should integrate end-to-end encryption (TLS 1.3) and role-based access control.

7. FUTURE WORK

A structured roadmap for TeleBot enhancement spans three development horizons. In the near term (6–12 months), priority improvements include: (i) replacement of the DHT11 with an MLX90614 non-contact infrared thermometer for clinical-grade body temperature measurement; (ii) integration of a MAX30102 SpO2/pulse oximetry module to add blood oxygen saturation monitoring; and (iii) implementation of TLS 1.3 encryption for the HTTP control interface and MQTT broker.

In the medium term (1–2 years), autonomous navigation using a Raspberry Pi 4 running ROS2 [16] with SLAM-based obstacle mapping [20] would remove the manual navigation burden. Integrating a cloud-based data persistence layer (AWS IoT Core or Google Firebase) would enable longitudinal patient records, trend analysis, and alert threshold personalisation [14]. A dedicated mobile application with a redesigned physician dashboard—including waveform displays for continuous BPM and temperature trend charts—would replace the rudimentary HTTP interface.

In the long term (2–5 years), the platform could evolve toward a multi-robot ward management system in which several TeleBots operate cooperatively under a central hospital information system,

automatically routing to patients based on predicted clinical deterioration scores derived from vital-sign time-series analysis [22,24,25]. Integration with electronic health record (EHR) systems would allow automated documentation of monitored parameters, reducing manual transcription burden. Socially assistive capabilities [11,17] such as voice interaction and emotional engagement could extend TeleBot's utility to psychiatric and rehabilitation wards.

8. CONCLUSION

This paper presented TeleBot, an IoT-enabled telepresence robot designed for real-time remote patient monitoring in hospital ward environments. The platform integrates wireless locomotion control, live MJPEG video streaming, DHT11 temperature sensing, PPG-based heart-rate monitoring, HC-SR04 ultrasonic obstacle avoidance, and Telegram-delivered emergency alerts within a single, cohesive mobile system built on affordable, commercially available components.

Experimental validation across all six functional subsystems yielded a composite pass rate of 98.3%, with particularly strong performance in emergency alert delivery (100% success, 2.7 s mean latency) and robot movement (100% command execution, <200 ms response). These results demonstrate that TeleBot can effectively substitute for periodic physical nurse rounds, extend continuous visual and physiological monitoring to under-resourced clinical settings, and provide rapid emergency escalation pathways—all at a platform cost below USD 60.

By demonstrating that clinical-grade remote monitoring functionality is achievable without expensive proprietary hardware, this work contributes a replicable, open-architecture template for affordable healthcare robotics in developing healthcare ecosystems. With the enhancements described in the Future Work section, TeleBot has clear potential to evolve into a comprehensive, intelligent, and ethically governed healthcare monitoring platform fit for large-scale deployment.

AUTHOR CONTRIBUTIONS

Shazana A: Conceptualisation, hardware integration, firmware development, manuscript drafting. Sasmita Vaissnavi P K: Sensor calibration, system testing, results analysis, manuscript review. Uvaniya V: Communication system design, Telegram integration, literature review, manuscript review. All authors approved the final submission.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest. The funders had no role in study design, data collection, analysis, interpretation, or manuscript preparation.

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